AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

M109
.Net Programming

Mid-term Examination
Spring Semester 2024/2025

Date: 8th April 2025

Number of Exam Pages: Five (including this cover sheet)	**Time Allowed:** 90 minutes

Instructions:

- Total Marks: **60**
- This exam consists of **3** parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are not allowed.

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Part 1: Multiple Choice Questions (MCQ) [10 marks]

1- **When you work with a _____, you can jump directly to any piece of data in the file without reading the data that comes before it.**

a. file object
b. binary file
c. direct access file
d. sequential access file

2- **The _____ file contains an application's start-up code which executes when the application runs.**

a. Main.cs
b. Code.cs
c. Program.cs
d. Source.cs

3- **When you create a new project using Visual Studio, it will be stored in a _____ at the location you specify on your computer's disk.**

a. Solution folder
b. Project foldeR
c. Solution file
d. System fileControls

4- **A _____ property that allows you to change the color of the tex tin a textbox.**

a. TextColor b. ForeColor c. FontColor d. Color

5- **Which property of a CheckBox control determines whether it is checked or unchecked?**

a. Selected b. Activated c. Enabled d. Checked

6- **..... Automatically deselecting any other items in the same group when a single item is selected.**

a. Multiple
b. Mutually inclusive
c. Mutually exclusive
d. Monographic

7- **In order to write data to a text file, you must create a file object using the _____ class.**

a. TextReader
b. TextWriter
c. StreamReader
d. StreamWriter

8- **Which of the following statements correctly creates a new instance of a Random object?**

a. Random rand = new Random();
b. Random rand = Random.New();
c. Random rand = Random.Create();
d. Random rand = new Random;

9- **A math expression involving a double and a(n) _____ is not allowed unless a cast operator is used to convert one of the operands.**

a. int b. double c. decimal d. Boolean

10- **Which of the following statements clears all items from a ListBox control named listBox1?**

a- listBox1.Items.Clear();
b- listBox1.Items.Empty();
c- listBox1.Clear();
d- listBox1.Items.RemoveAll()

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Part 2: Short Questions [20 marks]

Q1- Explain through code the exception handling in C#

Q2- Does the following code have an error? If yes, what is the error? If No, what is the output of the following program?

```
using System;
using System.Windows.Forms;

namespace WindowsFormsApp
{
    public partial class Form1 : Form
    {
        public Form1()
        {
            InitializeComponent();
            button1.Click += Button1_Click;
        }

        private void Button1_Click(object sender, EventArgs e)
        {
            string input = textBox1.Text;
            int number = int.Parse(input);
            int result = number * Math.PI;
            MessageBox.Show("The result is: " + result);
        }
    }
}
```

Q3- What is the output of the following program?

```
class Program
    {
        static void Main(string[] args)
        {
            Random random = new Random();

            for (int i = 0; i < 2; i++)
            {
                int randomNumber = random.Next();
                Console.WriteLine(randomNumber);
            }
        }
    }
```

Q4- Complete the following paragraph with the suitable terms.
[StreamReader, StreamWriter, FileStream, OpenFileDialog, SaveFileDialog, ReadToEnd, WriteLine]

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1. The ..... class is used to read data from a file in a Windows Forms application.
2. The ..... class is used to write data to a file in a Windows Forms application.
3. The ..... control allows the user to select a file to open in a Windows Forms application.
4. The ..... method of the StreamReader class reads the entire file as a string.

Q5- Draw the design of the form of the following code. Specify the output if the input is 3 .Trace each line not only the final output.

```
using System;
using System.Windows.Forms;

namespace WindowsFormsApp
{
    public partial class Form1 : Form
    {
        public Form1()
        {
            InitializeComponent();
            button1.Click += Button1_Click;
        }

        private void Button1_Click(object sender, EventArgs e)
        {
            string input = textBox1.Text;
            if (int.TryParse(input, out int number))
            {
                int factorial = 1;
                for (int i = 1; i <= number; i++)
                {
                    factorial *= i;
                }
                label1.Text = "Factorial: " + factorial;
            }
            else
            {
                MessageBox.Show("Invalid input. Please enter a
number.");
            }
        }
    }
}
```

Part 3: Problem Solving Questions [30 marks]

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Q1- You are tasked with creating a Windows Forms application that allows users to enter their age in a TextBox and display a message indicating whether they are eligible to vote (age >= 18) or not. [15 marks]
Requirements:

- The application should have a TextBox for entering age.
- There should be a "Check Eligibility" button to check if the user is eligible to vote.
- Display the result in a Label control.
- Ensure that the application handles invalid input (e.g., non-numeric values) gracefully using exception handling.

Q2- Write a C# program using Windows Forms to check if a given number is a prime number or not. The program should read input numbers from a file, check if they are prime numbers based on the specified criteria, and write the results to another file. [15 Marks]

A prime number is a number greater than 1 that has no divisors other than 1 and itself. For example, 2, 3, 5, 7, 11, etc., are prime numbers.

Your program should include the following controls:

- A Button labeled "Check Prime Numbers" to trigger the prime number checking functionality.
- A Label to display the status of the operation (e.g., "File processed successfully" or "Error occurred").

Additionally, your program should:

- Read input numbers from a file named "Numbers.txt" (one number per line).
- Check if each number is a prime number based on the criteria mentioned above.
- Write the results (number and prime number status) to another file named "NumbersType.txt".
- Handle file I/O operations correctly and provide appropriate feedback to the user.

End of questions

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